

COLLEGE CODE: 8203

COLLEGE: AVC COLLEGE OF ENGINEERING

DEPARTMENT: INFORMATION TECHNOLOGY

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ROLL NO: 23IT112

DATE:08/09/2025

Completed the project named as

Phase_1 TECHNOLOGY PROJECT

NAME: Online Quiz Application

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PROBLEM STATEMENT:

In many schools, colleges, and training centers, quizzes are still conducted in the traditional paper-based method. Teachers spend time preparing question papers, distributing them, collecting answer sheets, and then manually correcting them. This process is slow and often causes delays in giving results to students.

Manual correction can also lead to errors and may not always be fair. Students do not get instant feedback, which reduces the learning impact. In today's online learning environment, quick evaluation and feedback are very important.

There are some online quiz applications available, but many of them are costly or do not provide enough flexibility for customization. Small institutions and individuals face difficulties in using such tools effectively.

Because of these issues, there is a need for a simple and efficient Online Quiz Application. The system should allow quiz questions to be stored in a database, provide an easy way for students to attempt quizzes, automatically evaluate answers, and generate results instantly. At the same time, it should allow admins to add, edit, and update quiz questions without difficulty.

USERS AND STAKEHOLDERS:

Users

Students/Participants – Take quizzes, answer questions, and view their scores.

Admins/Teachers – Add, update, or delete quiz questions and view results.

Stakeholders

Students/Users – Benefit by practicing and improving knowledge.

Teachers/Admins – Manage quizzes and monitor performance.

Institutions/Organizations – Conduct online tests and assessments easily. **Developers** – Build, maintain, and improve the system.

USER STORIES:

1. As a Student/User

I want to view available quizzes so that I can choose and attempt them.

I want to answer multiple-choice questions so that I can test my knowledge.

I want to see my score immediately after submitting the quiz so that I know my performance.

I want to review correct answers after submission so that I can learn from my mistakes.

I want my past quiz history to be saved (optional) so that I can track my progress.

2. As an Admin/Teacher

I want to add new quiz questions so that students can attempt updated quizzes.

I want to update or delete existing questions so that quizzes stay correct and relevant.

I want to view student results so that I can analyze performance.

3. As an Institution/Organization

I want to conduct online assessments so that tests can be managed easily.

I want to generate reports from quiz data so that student progress can be evaluated.

MVP FEATURES:

User Login/Signup (optional basic auth) – allow users to access the quiz.

Take Quiz – user can attempt multiple choice quizzes.

Quiz Questions from DB – questions/options fetched from MongoDB.

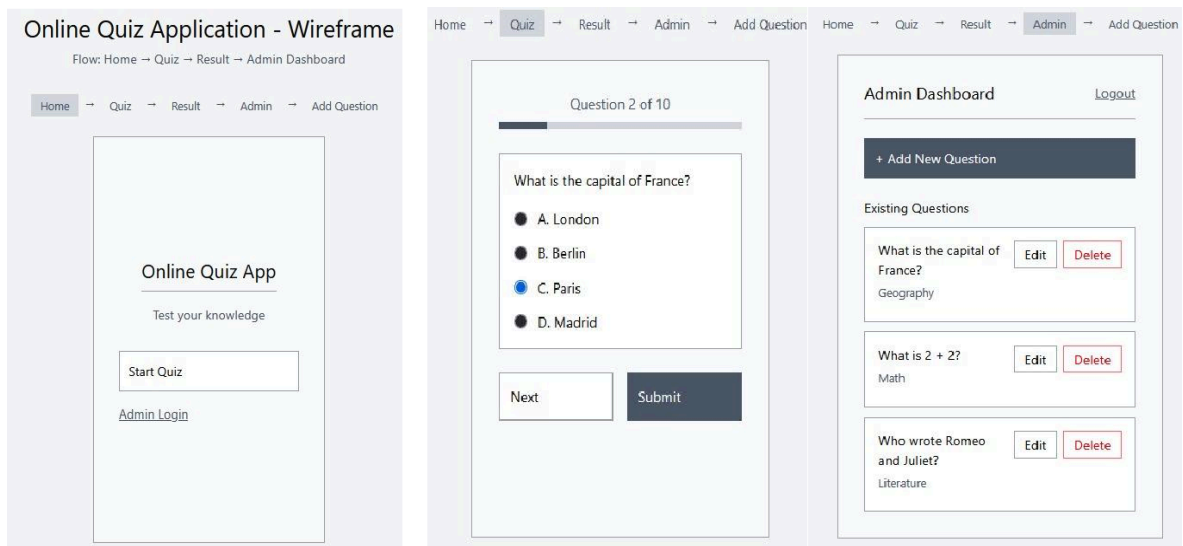
Submit Answers – user submits answers.

Score Calculation – backend checks answers and returns score.

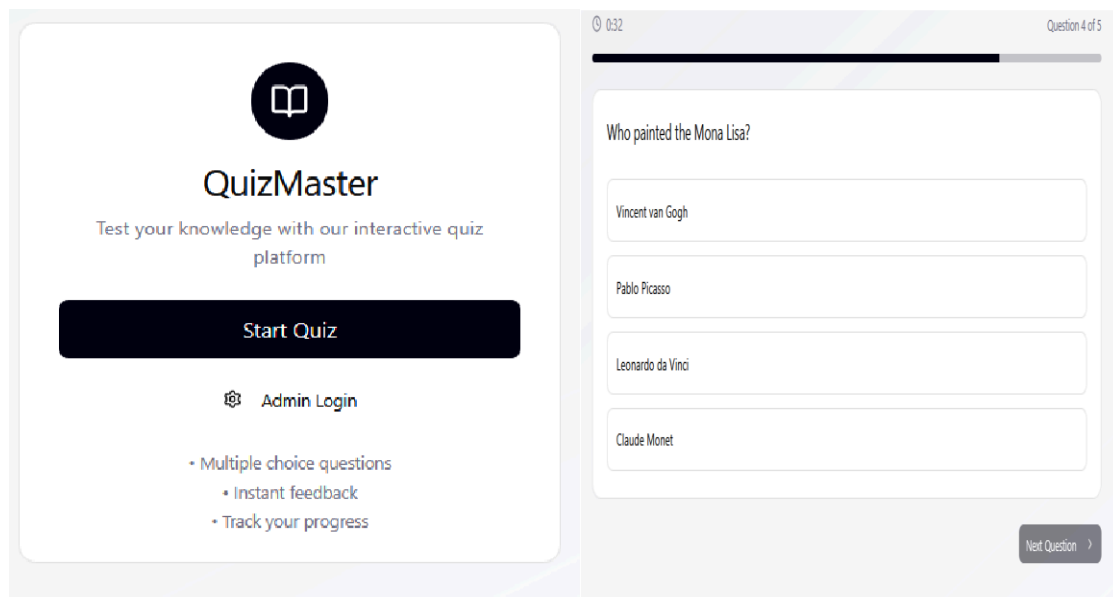
Result Display – show user's score after quiz.

Basic Admin Panel – add/update quiz questions.

WIREFRAMES:



USER INTERFACE:



ACCEPTANCE CRITERIA:

1. Quiz Functionality

- Users must be able to view and attempt quizzes with multiple-choice questions.
- Each question must display 4 options with only one correct answer.
- Users must be able to navigate through questions and submit answers.
- The system must calculate and display the score immediately after submission.

2. Admin Features

- Admins must be able to add new quiz questions with options and correct answers.
- Admins must be able to update or delete existing quiz questions.
- Admin actions must update the database in real time.

3. Result & History

- The result must clearly show the total score and correct/incorrect answers.
- The system should save quiz history for each user.

4. System Behavior

- APIs must fetch questions and validate answers correctly.
- The application must work on standard web browsers.
- Data must be stored and retrieved from MongoDB without errors.